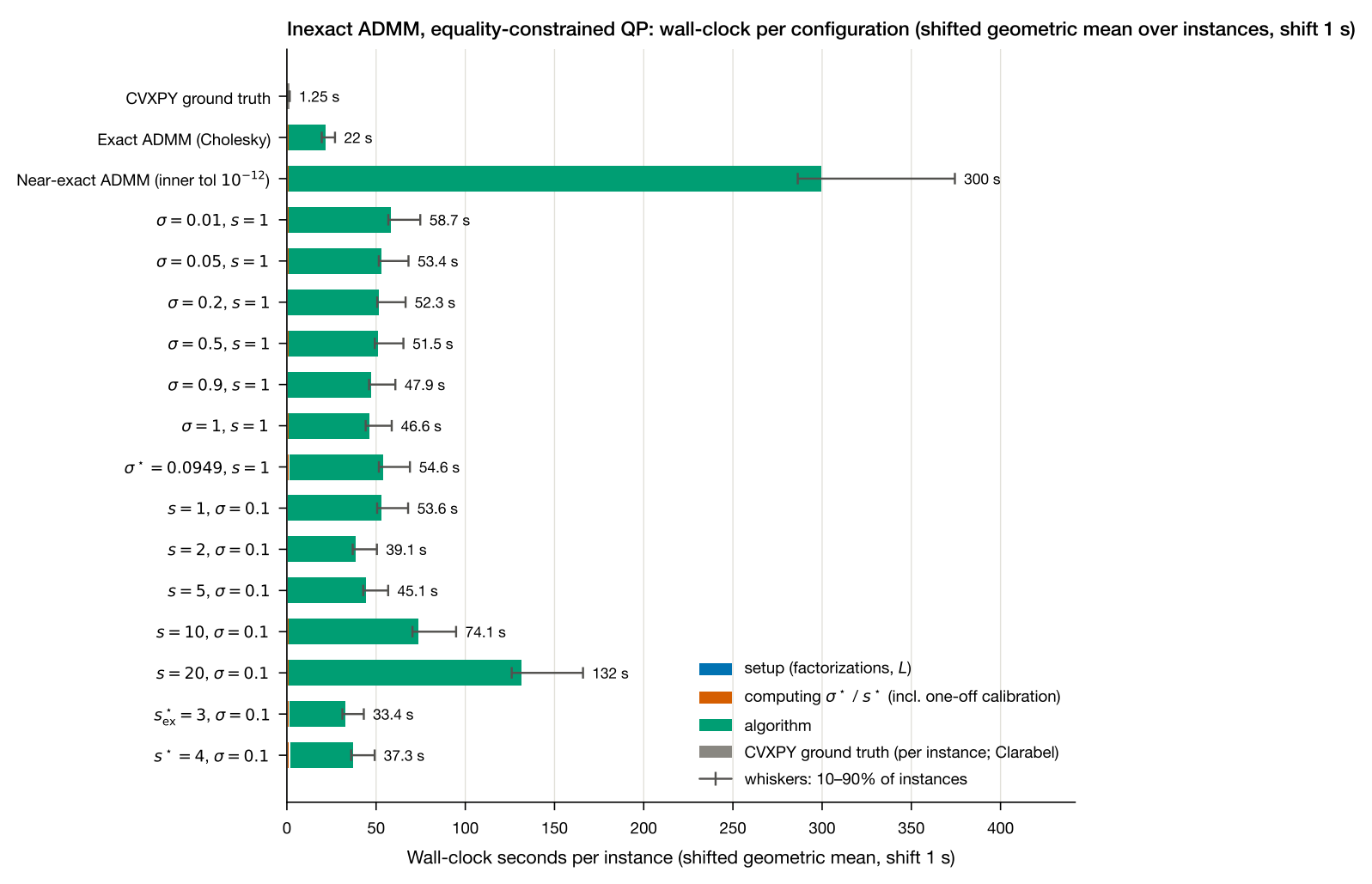
Inexact ADMM, equality-constrained QP: wall-clock per configuration (shifted geometric mean over instances, shift 1 s)
CVXPY ground truth
1.25 s
Exact ADMM (Cholesky)
22 s
Near-exact ADMM (inner tol 10^{-12})
300 s
$\sigma = 0.01, s = 1$
58.7 s
$\sigma = 0.05, s = 1$
53.4 s
$\sigma = 0.2, s = 1$
52.3 s
$\sigma = 0.5, s = 1$
51.5 s
$\sigma = 0.9, s = 1$
47.9 s
$\sigma = 1, s = 1$
46.6 s
$\sigma^\star = 0.0949, s = 1$
54.6 s
$s = 1, \sigma = 0.1$
53.6 s
$s = 2, \sigma = 0.1$
39.1 s
$s = 5, \sigma = 0.1$
45.1 s
$s = 10, \sigma = 0.1$
74.1 s
$s = 20, \sigma = 0.1$
132 s
$s^\star_{\text{ex}} = 3, \sigma = 0.1$
33.4 s
$s^\star = 4, \sigma = 0.1$
37.3 s
setup (factorizations, L)
computing $\sigma^\star$ / $s^\star$ (incl. one-off calibration)
algorithm
CVXPY ground truth (per instance; Clarabel)
whiskers: 10–90% of instances
0
50
100
150
200
250
300
350
400
Wall-clock seconds per instance (shifted geometric mean, shift 1 s)